

Active Inference Institute ~ 2022 Quarterly Roundtable #3

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[Music] thank you welcome everyone this is the active inference Institute quarterly Round Table number three in 2022 it is September 28 2022. welcome to the active inference Institute the active inference Institute is dedicated to learning researching and applying active inference you can find us at some of the links here on this slide this is recorded in an archived live stream so please provide us with feedback so we can be improving our work all backgrounds and perspectives are welcome and we'll be following video etiquette for live streams today in the quarterly live stream number three we are going to have a three broad sections they won't be too long there will be a lot of time if you want to add any other notes in the live chat or maybe even if anybody wants to um say hi in live chat and join the uh live stream itself we're going to first have Institute scale updates we'll talk about some changes in development at The Institute level we'll then talk about projects and the Education and Research Endeavors that we've been involved in and that have been developing over this quarter and then we'll close with some next steps and affordances for getting involved let's go first to the active inference Institute aii updates at The Institute scale all right I'll just give a some overview and then Alex or Yvonne feel free to add more we're continuing to work with fried Frank in a pro bono capacity to help with the legal side of formalizing our incorporation as a non-profit we are formalized in terms of what members we have our five founding members and some of our governance structures currently we are on the stage of selecting and electing the board of directors so at this point at the end of September 2022 we are extremely actively seeking self-nominations as well as suggestions for people who might like to serve in the board of directors especially with expertise or experience serving on board of directors for non-profits and any other relevant experience related to governance Financial non-profit Education and Research following the selection of the board of directors the board will then select officers who are in uh more of a direct operations role at The Institute and we've also joined the Delaware Alliance for non-profit advancement Dana to provide us with access to non-profit resources

and support Alex or Yvonne want to give any comments if you're talking Alex you're muted but I see yellow okay okay carrying forward We are continuing to develop our Institute branding first we are working with some external expertise on some logo and brand redesigns for The this is really exciting and more will be announced when the work here is ready to be shown so enjoy the black white and red and the current fonts and layouts that we have while you can they'll still be there they'll still be around and we've also updated our website with a restructured short description of the Institute which I'll just read now and then Alex riven please make any extra comment or provide any context applying active inference we provide avenues for connection and integration with broad audiences and disciplines the active inference Institute provides a setting for people to Aid each other in pursuit of a better understanding of active inference The Institute organizes education research and communication to advance the progress and public awareness of Frontier knowledge in active inference and closely related topics we structure our strategy with the building blocks of participatory open science accessibility and service to our epistemic community wanna yes yeah sure uh as a as a institution we are educated to um having understanding or in a broad audience about active inference what is it why how we can use it and as we know that active inference is applicable for a lot of fields and domains we are open to the community from ho ho all of those fields and their names thanks yeah agreed that one of the best ways for a connection to be made between active inference and some other field is kind of the micro engagement when people from that field get involved and get involved in learning and then they'll see all these affordances for connecting active inference theoretically and in practice to their area it's not like there's just a tree and the connections are ready to be pulled at First Sight it's takes a little integration and including different communities in the conversation so that's what we're happy to be building and working on and also in a participatory way where people's feedback is really welcome and people's engagement is essential so we'll continue with a few more notes on the Institute scale and then we'll talk more about projects that are happening at the Institute all of which people can get involved in if they would like so just a structural note uh as a summary the focus and functions of our Institute are research education Outreach and engagement and methods and these Foci are playing out across multiple scales there's the participant scale which is the human and their engagement with the there's the Institute scale which is the system of interest that we're able to enact policy in at The Institute level and there's our broader epistemic Community which includes the really exciting and growing active inference community and all the amazing developments happening in that area

as well as more broadly and um more distally many adjacencies that have been identified or have not been identified ranging from neuroscience and medicine to machine learning computer science ecology robotics all these amazing areas we have discussions around and we have two organizational units that are kind of like departments they are organizational constructions that help us coordinate projects and efforts and allocate attention around our two primary missions in Foci which are research and education so Azure active is the organizational unit that hosts education projects whereas the Research Unit hosts research projects and we uh we joked about having a kind of competition or an opening of window for renamings but as anyone who has watched these roundtables know we show up on the day of with what we know that day so your engagement or suggestions and ideas are always welcome and that's how we update that's how we learn but that's just reflecting some of our structural learning and updating that's the Institute level anything else to add here Alex driven or we'll continue on to the project scale all right let's talk first about education this is going to be education at the Active inference Institute and project updates over Q3 2022. first the active inference textbook group so we've been having two concurrent cohorts of the active inference textbook group the first cohort of the textbook group began in May 2022 and over the following three months they completed the first five chapters of the textbook and now that first cohort the Intrepid adventurers the early foragers are continuing on in chapters six through ten which is where the textbook turns towards model building and recipes for model construction which has been really exciting and we also in September of 2022 began the second cohort of the active inference textbook group so this is some people who are rejoining from the first cohort as well as many many new people joining in and it's another concurrent cohort that's starting from the beginning of the book so this is the kind of minimal templating of a continuous delivery textbook group format where as more and more people are interested and curious to learn about active inference and increase their familiarity and competency in engaging and positive social environments we will be able to have the familiarity of people who have taken the cohort experience of the textbook group before to then return for example as facilitators or in other roles and therefore scale The Continuous delivery of textbook group experiences around the par pazulo and friston 2022 textbook but also around other texts that may be arising where as we work as a group we're in a shared epistemic Niche and we provide a whole host of value-added services that deeply enrich your learning so if you're currently spending one or more hours per week or month curious about active inference especially curious about this textbook your learning and your contributions will be leveraged if you join

a textbook group cohort we curate and we compile and address questions that we're having relating to different parts of the textbook and we've created just an amazing repository of questions that grows every day and they contain questions that are confirmatory and basic what does this word mean here or what do you think this means all the way on up to speculative and research oriented or application oriented questions so questions are really important part of learning and they're a really important part of the textbook group and also through being in the textbook group one becomes more familiar with a variety of ontology scaffolded learning approaches that we use in the active inference Institute related to our work from the very beginning with the active inference ontology so for example here's equation 2.5 with a variational free energy and here we have just a screenshot of what The annotation looks like in the shared knowledge resource in the coda for those who are learning and discussing equation 2.5 and there's a whole host of other information questions that people are having and what does this mean and what are the relations to other areas but this kind of a learning environment allows people who have different backgrounds in math or the formalisms or in some of these other theoretical domains that come into play in relationship with active inference to have a really exciting space where we're all learning a ton so it's been awesome to have two concurrent cohorts going in the textbook group and we're looking forward to in the close of 2022 to have the first cohort complete the book and then have a second cohort get up to the midpoint and then in the beginning of 2023 we will likely continue with the second cohort chapter 6 through 10 and begin a third cohort and see where things are at in the active inference world then any comments or thoughts on the textbook group uh textbook group is very valuable tradition for education and when you learn something and you get feedback from your peers you have more understanding and in our groups we are providing a lot of useful instruments that help to understand not so clear Theory as active inference awesome and one kind of note or seed for the future is the textbook is open source which is awesome it increases the accessibility and downloadability but it also offers some incredible future opportunities in terms of remixing the book transposing the content into GitHub re-rendering translating doing some amazing secondary work with a textbook and again seeing that as templating for future versions or other textbooks so that's the textbook group onwards we go on once once again the main goal of this activity is to help understand our main idea and we plan to do this activity on a regular basis and you can join us wherever you want and wherever you are ready thank you Yvonne very true all right in July 31st 2022 we had our second applied active inference Symposium on robotics there were two live streams of about four

hours each and there were a handful of co-organizers from within the Institute and from outside the Institute we had a whole range of formats presentations roundtables and discussions the two sessions are shown here and I'll uh when somebody else is talking make sure to post the links in the YouTube live chat so people can go check them out or they're available just on our live stream Coda page there were about nine presentations of around 20 to 30 minutes each people talking about specific applications of active inference so if you're wondering where is active inference being applied or how can active inference be applied this Symposium gives nine case studies and settings from the robotics domain where active inference is being applied today and tomorrow and at the end of each session we also had a round table discussion that was really interactive and connected the dots and provided some context beyond the more research-oriented presentations that were happening for the bulk of the presentations so for those who are interested in applications of active inference or in robotics this was a really exciting event and this is also we hope a tradition to continue applied active inference symposia where people who want to co-organize or present on their applications in Knowledge Management or in medicine or in ecology we can convene and coordinate around the symposia and have different formats so we're currently working to transcribe the products of this Symposium and make them citable proceedings any thoughts you have on this you fellows I think that robotics in terms of Applied field is one of the most straightforward application of active inference and possibly and future providing real breakthroughs in results of robots that will be used active influences from work for where Robotics cognitions and so on and as we said from web communication there are different groups of researchers all over the world we are interested in communication they are interested in in presence in results and hopefully more people next years will join in with robotics active influence robotics movement and some dates in some days some close future we'll see really great application and like autonomous robots based on active inference awesome thank you all right speaking of the active inference Journal so the goal of the active inference Journal is to increase the accessibility and the indexability citability and utility of live streams and various audio visual material produced in and around the active inference Institute So currently we've been working on around four specific streams and events to improve and template our process that we're hoping to build and scale on further what we do in the journal is we take the outcomes of the live stream the audio visuals of the live stream and we get a primary artificial intelligence AI generated transcript and then engage in a sequence of enrichments editing and curations so we edit the transcripts by listening carefully to the audio and making sure that the

speaker's sense is to the extent possible captured and then we also develop for example appendices and other scholarship around the document so when somebody mentions a paper or when somebody mentions a person those kinds of information can be included and the discourse can be connected to the active inference ontology in cases where the terms are being used and again that will enable more accessible searching in the language of the live stream for example people have asked questions that somebody might want to query again or might want to hear someone's response at a given time too and cite that or just learn from it and then also it's really important to have such standardized pipelines for transcription in order to enable translation into different languages ways that you can get involved in the journal if you're interested or curious in this area whether you have editing or proofreading experience or not there are many affordances to edit transcripts so listening carefully to the audio and providing suggested edits on texts and also we're especially interested to research and develop methodologies for the journal process so anyone familiar or curious about AI assisted transcription and translation modular Publications decentralized science and DSI the usage of git as a versioning system any of these areas we would welcome your engagement in the journal project because that's the direction that we want to head and that is what is going to unlock some of the potential of the 200 plus live streams that we've already carried out and the many hundreds that we'll carry out in the future it will enable these past and future live streams to be able to be searched cited translated and used more broadly so if you want to join the meetings for the journal project they're on Mondays at 13 UTC in our Discord so a few next steps I mentioned before but I'll just give a quick shout out to um at carpathy who recently for the Lex Friedman podcast spun up a site using the open AI whisper platform and made transcripts for all Lex Friedman episodes and so we imagine something similar because in our Coda for the live streams we have 200 plus Journal discussions research presentations we are ready to be open AI whispered or other Pipelines so if that is something you're curious about or carpathy or others if you want to come get involved you're absolutely welcome there's a lot of learning and accessibility and rigor that can be improved with these kinds of rapidly emerging text engineering developments any other comments you want to make on active Journal yes and this 200 plus videos there are a lot of a lot of useful information that's very valuable and if it will be extracted it will move forward while we or how we understand it both episodes totally agreed there's ways of saying and conversing in the live streams that certainly are clearer restatements of and developments beyond what is in the written literature so especially as we move towards a multimedia future and a more participatory and inclusive

science sector it's important that we have the systems in place with low human bandwidth to be able to improve and distribute scholarship on these topics All Right Moving On to the live streams themselves so we had uh one month off in July from live streams where we just chilled read other papers not these papers um and then we've had a bunch of great live streams so far in Q3 we had guest streams with Brett Anderson with Ronan tamari at all and with John verveiki and we discussed papers in the live stream series on an active cognition abductive logic and active inference and active inference as ecological communication we also have a wide range of really interesting live streams coming up including on Bayesian mechanics on neural networks on all kinds of areas of active inference application and a lot of really exciting theoretical developments also so if you want to be involved in live streams in the YouTube live chat you'll find the past and upcoming streams so one way to get involved is to watch listen share comment and be involved that way if you want to play a role for example speaking on a live stream or helping contribute to the preparation around the live stream that can also be incredibly important as well as making connections and recommendations for whom we should invite so that we can have a really powerful and diverse range of guest streams and papers selected for the live streams so live streams have continued in their educational function and just wanted to highlight um each one of these projects there have been many people to mention and I hope that we can by next quarterly lab meeting have some clearer Pathways for making those clear but just wanted to it in bulk acknowledge everybody who contributes to the weekly meetings for live stream preparation and organizing and those who join the stream as well as those who help prepare content and coordinate questions for the live stream in the background and may not be on the video themselves so that's the live streams we continue on there any other comments just once again uh one obvious think the more you the more you uh involved in any any of any our activity and live streams is uh extremely extremely useful for understanding and to see the way of application affect this inference the more you're engaged the more you understand this topic great point and sometimes just scanning the titles of the papers that we discussed in the 50 papers that we have discussed almost 50 by the end of the year these are curated selections usually with the authors joining and almost always with a DOT zero video where people who cared made background in context to make it as accessible as possible so this is a unique Learning Resource if you have a regime of attention and availability or capacity for these topics or you're curious and think you might it's going to be a really unique Learning Resource for you to check out our past videos and then if you're curious about media production communication you want to explore short

formats you'd like to facilitate a discussion in a non-english language you want to make an animation video there's so many affordances that we can support So if you're curious about those areas just get in touch okay continuing on education internal systems thinking ongoing activity uh Yvonne let's I'll let you describe it thanks yes so we also have not actually related the active inference our activity and this August we launched uh four weeks session for assistant thinking course part of it that related to the role Concepts uh we for four weeks we can't cover the whole the course foreign just a second we can't cover the whole course but four weeks for understanding the role concept is uh fair enough time possibly also also run this October next cohort and for previous one we have um good feedback for um from our participants could use uh could you show next slide please thank you next this October we start a new cohort and here is a link for participation will also it will also take four weeks and during this week's we consider uh concepts from system thinking course yeah and I will apply these Concepts to the ER Pro to your projects or maybe for even live projects and next one we we have this uh activity with our partner Institute for augmented intelligence in entrepreneurship engineering and management and this fall The eam Institute launched for courses now these courses are available for free and we also encourage you to discount discourses for sure you will find a lot of a lot of useful information also related to the active inference as a evolutionary theory awesome yes so this system's engineering and systems thinking activity helps us learn about roles and how to apply them in projects and we are applying them in active inference Institute projects so it also upskills participants and gets them ready to take on more involved and responsible roles in our Institute and it's all a co-evolutionary dance as The Institute formalizes and gains resources and establishes itself in the niche then such roles will also be the ones that we can support all right any other comments onto the research so the previous updates were from education just to Briefly summarize textbook group two ongoing cohorts active inference robotics Symposium with eight hours of presentations and active inference Journal working towards accessibility and indexability of all of our speech products live streams continuing and systems thinking education activity ongoing open to all of these projects your involvement on to the research so it's been a great quarter for research first we'd like to highlight our active block prints development active block prints is an open source package that is implementing classic and novel active inference models within the system simulation framework of cadcat so we have a few updates on active block prints first Jakob smickle presented a poster at the iy 2022 third international workshop on octave inference so congratulations Jacob for this awesome poster presenting it in person

and to all others who have been contributing to development on octoblock prints it's um really great to see the development of this project and also a lot of other work in the active inference community so please check out iy if you're curious to see what is Cutting Edge in active inference research they really did a great job in getting some of the best projects in the space um also active blockference has received um some funding through the git coin gr15 round that just completed so our git coin project was active block prints an open source package for cognitive modeling in web 3. and uh we specifically framed how we could utilize cognitive modeling approaches integrating active inference and cadcad for cognitive modeling in web3 ecosystems so big appreciation to everyone who provided even one dollar to this funding opportunity and in several weeks we'll know the final amount that we've received after the quadratic funding algorithms have all been applied but suffice to say that we have some money in our tank to develop active block prints so there are many ways and areas people can get applied in there are weekly meetings there are weekly meetings at 12 and 17 UTC on Wednesday in the Discord and also we discussed this um with Jacob today Mondays at 18 UTC is going to be a code development specific session so if you want to get a little bit more Technical and collaborate with Jacob and others on the cad CAD active inference modeling then Mondays 18 UTC is a great synchronous time at 12 and 17 UTC on Wednesdays those are slightly more open sessions where we can also split off and have a technical development but we talk about a whole range of areas so I'm going to mention some areas and then tasks and how we're going to use some of our funding to accelerate development these are just showing a few of the areas where we believe cognitive modeling could be used or might be important or relevant to consider ranging from a range of social settings education governance legal online work stigmergy so on to a host of web 3 cases so Dows decentralized Finance d-school DSi ethereum a lot of other areas in web 3 and then also uh on the ground with the ants in terms of developing the active infra ant model for ant colony foraging and ultimately larger task allocation questions these are all areas where we see the active block runs package as being able to address them eventually and what it takes to address an active lock for its model in one of these cases and settings is for someone with domain expertise or knowledge to provide the Bridge Building from one side and then the package to develop the generalized capacities from the other side so we're working from both ends to enable really powerful domain-specific applications of active inference while building modular structure of the active block fronts package so that those functionalities can be used in other settings for example a navigation setting that might arise from our work on Ant foraging might be used in a different

context in another non-ant setting so that's a little bit about how we approach the development of the package we think about the functionalities of the package modulars as modules that can be invoked and functions that can be called and then we develop different settings for cognitive modeling there are a range of tasks that people can get involved in ranging from cadcat and active inference technical tasks to a whole host of package level and documentation and Community engagement tasks so if this sounds like a cool direction or you would just like to learn more then please join a synchronous meeting or get in touch another way because this is a super exciting area of development we hope that active block prints can be a really important package in this area and so come be involved with the funding what we're going to be experimenting with in the coming weeks and months is a task based payment scheme so individuals or small teams will propose how much funding they would like to receive for what specific tasks and there's a form that will be filled out and so we can show you that if you join synchronously completing that form will raise your task requests to consideration and then if the task is approved and you complete it as laid out you will be paid the amount that you requested so if funding ever felt like it was a rate limiting step for your involvement in learning or applying active inference and contributing to open source public goods then consider it differently now because we can support this type of work and we also see some really exciting opportunities for people for example to propose tasks around applying to other grants or to including more developers or more documentors or people who want to make active block runs live streams there's a million areas where people can develop here so join the weekly meetings is the best way to get involved there any other comments on blockfronts last research update this is not presenting all the research happening in The Institute this is just to kind of raise a few projects up we have had a project on cognitive modeling Frameworks and their comparisons the goal of this project is to understand similarities and differences with active inference and other cognitive modeling Frameworks so the approach that we're taking here is to review and assess different in terms of their features their functions their elements their areas of application and so on and that will help us understand similarities and differences amongst Frameworks as well as more formally and historically integrate active inference with other models so this is one of a research task that could open up quite broad Vistas for application anyone who's curious or engaged in cognitive modeling would be welcome to get involved with this project and one can even see the pathway here kind of like the epistemic chaining example in pi mdp if someone identifies that there's a common cognitive modeling framework that has features ABC and is used in domain

f then domain F becomes an area of application for active inference if functions ABC can be implemented better in active inference than the initial framework so that kind of work on the landscape of cognitive modeling is a really important precursor to understanding where and how and with an emphasis on which functionalities it even makes sense to talk about applying active inference and we hope and expect and prefer that there will be active inference applications that haven't been within the range of traditionally considered cognitive settings for example I'm not sure how often in multi-scale ecology cognitive modeling Frameworks are used as a simulation tool now we know that active inference is being applied in those areas but there are also areas where cognitive modeling is already happening and it may or may not be using active inference so those represent adjacencies whereby developing the connection between how they're doing it today and how it could be done people will be opening up really large areas for active inference application and also massive areas of attention and support so this is kind of a cool Direction any notes on this or anything else on all right let's just go to some last steps and close if anyone in the live chat has any questions they're welcome to add them so next steps for The Institute well on the engagement and Outreach Frontier we continue to grow and develop we host most of our Communications at our Discord which has over 450 members now uh our Twitter as well as our podcast and video channels are all growing in terms of their niche so if you support us on any of these platforms and can share or just even tell one person that can be quite helpful and always provide us with feedback on how we can improve our environmenting our engagements with external interfaces and then just to provide a summary of some steps and directions for how we go from Q3 on Beyond to Q4 any of these areas if you are interested or curious please be involved the earlier and the more involved simply the better and the higher leverage point that is for the future We are continuing education and research it's the main focus of the active inference Institute and we will always continue education and research we're also during these months undergoing a really creative and unique organizational development opportunity for example around assembling the board of directors and some of the early decisions around governance and organizational structure so if you have input or want to be more involved here as well otherwise just to know that it's happening we are looking towards several directions in the end of 2022 and to really come into in a new way in 2023 we're looking towards increased participation in projects and learning groups as well as the development of an internship program that will help structure your engagements with a mentor and in a Project Specific way so that you can stay engaged and also have something clearer to point to about your

engagement with the Institute We are continuing to as we finish the first cohort of the textbook group think more and more about active inference course development so we'd encourage anyone who wants to partner or be involved in that area to get in touch in the journal project and in a slightly different way in the textbook group we're working towards the captioning the subtitles and language translations where possible and especially with our new organizational structure we're looking towards funding opportunities and Partnerships so if you have any thoughts or capacities or feedback in these areas they're all super important areas for The Institute going forward and it would be really appreciated as we are a participatory open Science Institute and only recently received our first small amount of funding through the get coin grant for Block prints to explore new avenues of partnership funding service and beyond that we have not done so what hasn't happened yet is not too strong of a prior on what we may do in the future if that makes sense we'll be doing different things and I hope that you can be a part of it because it grows and changes in co-engagement when and only when people get engaged that's interactionism this is not instructionism so if you have any excitement around these topics or any others that you would want to have on here then getting in contact with us will be a great first step join our Discord join any of the synchronous meetings email us and we can start a conversation that way well that is all for this round table it was a short quarter it was only like few months from our previous live stream time went by quite fast and a lot did happen so once again next next for the promises to the uh more active and as you understood they are very open to participation so see you in the class see you on our our synchronous meetings in Discord subscribers and share yes Alex any last comments um I can only repeat that the participation of people who will join us in future is the main focus for us and everything we what we are doing is related to help others to learn about active influence to learn active inference and find ways to apply it and we are on open and endedness path and actually we we don't know what will be in in three months from here but we will adopt and provide the possible affordances for hours to join and learn so welcome indeed welcome to those who are and haven't joined yet and a big appreciation again for everyone who's contributed through the Institute to make things how they are today so thanks Yvonne and Alex see y'all later thank you thank you